

SIMATS ENGINEERING

ASSESSMENT TOOL 2

Course Name: Artificial Intelligence

Course Code: CSA17

Course Outcome (CO4): Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

Duration: 1 hour

Total Marks: 40

Annexure A — Article Review: Code of Conduct

I M. Arshini(192424322) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Date: 31-08-2026

ARTICLE REVIEW

Application of Machine Learning Models Based on Decision Trees for Classifying COVID-19 Mortality

Task 1: Article Summary

(a) Full Citation, Domain and ML Problem

Moslehi, S., Rabiei, N., Soltanian, A. R., & Mamani, M. (2022). Application of machine learning models based on decision trees in classifying the factors affecting mortality of COVID-19 patients in Hamadan, Iran. BMC Medical Informatics and Decision Making, 22, Article 192. <https://doi.org/10.1186/s12911-022-01939-x>

Journal	BMC Medical Informatics and Decision Making, Volume 22, Article 192
Publication Date	24 July 2022 (Open Access, peer-reviewed)
DOI	10.1186/s12911-022-01939-x
Domain / Application	Healthcare analytics — clinical decision support for COVID-19 mortality risk classification
ML Problem	Binary classification of COVID-19 patients as recovered or deceased using interpretable decision-tree models

(b) Objective Summary and Research Gap

The study focuses on building an interpretable machine-learning approach for identifying patients at higher risk of death from COVID-19. The researchers used medical-record data from 2,470 confirmed COVID-19 patients admitted to two hospitals in Hamadan, Iran. Instead of feeding all available variables directly into one classifier, they first used Random Forest feature importance to screen 25 demographic, clinical, and laboratory variables. Features with relative importance above 6% were retained, resulting in age, gender, blood urea nitrogen (BUN), partial thromboplastin time (PTT), serum glutamic-oxaloacetic transaminase (SGOT), and erythrocyte sedimentation rate (ESR) as the principal predictors. The selected variables were then evaluated using four decision-tree approaches: Logistic Model Tree (LMT), C4.5, C5.0, and CART. The research gap identified by the authors was that earlier COVID-19 mortality studies often reported predictive performance without clearly interpreting how important clinical variables were related to mortality. The authors addressed this gap by combining feature selection, repeated train-test evaluation, and an interpretable CART flowchart that converts the model into understandable clinical decision rules. This makes the work useful not only for prediction, but also for prioritising clinically meaningful risk factors during time-sensitive decision making.

TASK 2: METHODOLOGY ANALYSIS

(a) ML Techniques and Dataset Preparation

The methodology combines ensemble-based feature selection with interpretable decision-tree classification. Random Forest was first used to rank the importance of 25 demographic, clinical, and laboratory variables. The authors generated 1,000 trees and used the Gini criterion to calculate feature importance; variables above the 6% importance threshold were selected. Four classifiers were then compared: Logistic Model Tree (LMT), C4.5, C5.0, and Classification and Regression Tree (CART). LMT combines tree splitting with logistic models at nodes, while C4.5 and C5.0 use information-gain-based tree construction and pruning. CART produces binary splits using criteria such as Gini impurity or entropy and can be pruned for simpler decision rules.

The dataset contained 2,470 RT-PCR-positive COVID-19 patients admitted to Sina and Besat hospitals in Hamadan, Iran, between February 2020 and July 2021. The original variables covered demographic factors such as age, sex, marital status, smoking and comorbidities, together with laboratory measurements including ESR, BUN, blood sugar, CPK, SGOT, PTT, platelets, sodium, LDH and PMN. The outcome was binary: recovered (0) or died (1), with 1,853 recovered patients and 617 deaths. Records containing missing feature values or outlier/illogical values were removed after the missingness mechanism was examined. For model evaluation, the dataset was repeatedly divided into training and test subsets ten times, and mean recall, accuracy and F1-score were calculated.

(b) Mapping to CO4 and Methodological Strength / Limitation

The article maps strongly to CO4 through Decision Trees and Inductive Learning. LMT, C4.5, C5.0 and CART are direct examples of decision-tree algorithms, while Random Forest demonstrates how tree ensembles can support feature selection. The supervised training process also represents inductive learning because decision rules are learned from labelled patient records and then applied to unseen cases. A major methodological strength is interpretability: the final CART model converts statistical relationships into a visible tree structure that clinicians can inspect. Another strength is repeated train-test evaluation, which reduces dependence on a single random split. A limitation is the restricted data source. The patients came from only two hospitals in one region of Iran, and the authors note that some laboratory variables such as ALP and ALT/SGPT could not be included because of missing information. Therefore, external generalisation to other hospitals, populations, and healthcare systems remains uncertain.

The strongest part of the approach is the combination of feature ranking and transparent classification. Rather than presenting a black-box prediction alone, the researchers identify the variables that contribute most to mortality and then build an interpretable tree. However, the feature-selection threshold of 6% is data-driven and may not produce the same selected variables in another population. In addition, repeated random train-test splits are useful but are not equivalent to independent external validation. The imbalance between recovered and deceased patients also means accuracy should not be interpreted alone; recall and F1-score are especially important because missing a high-risk patient has a different consequence from incorrectly flagging a lower-risk patient.

Task 3: Results & Critical Evaluation

(a) Key Results

CART was the best-performing model on the reported evaluation criteria. On the test data, CART achieved a recall of 0.955, F1-score of 0.8681 and accuracy of 0.7824. For the complete dataset, the reported CART values were recall 0.9385, F1-score 0.8667 and accuracy 0.7834, with an AUC of 0.795. The authors also reported training accuracy values of 76.89% for LMT, 77.70% for C4.5, 76.89% for C5.0 and 78.24% for CART, with CART having the highest training F1-score at 86.81%. The results indicate that CART offered the best balance among the compared tree models, particularly in identifying patients who died.

Comparative performance of the decision-tree models reported in the article:

Model	Type (CO4)	Test Recall	Test F1-Score	Test Accuracy	AUC
LMT	Decision Tree + Statistical Model	0.9261	0.8573	0.7689	—
C4.5	Decision Tree	0.9189	0.8608	0.7770	—
C5.0	Decision Tree	0.9261	0.8574	0.7689	—
CART	Decision Tree	0.9550	0.8681	0.7824	0.795

(b) Critical Evaluation

The reported results are plausible for a clinical tabular dataset, but accuracy around 78% shows that the model should not be interpreted as a standalone diagnostic or treatment system. A useful observation is that CART achieved a high recall of 0.955 on the test set while its accuracy remained 0.7824, illustrating why several metrics are necessary for mortality classification. There are potential sources of bias. The study is retrospective and based on only two hospitals in Hamadan, so hospital-specific treatment patterns or patient characteristics may influence the learned rules. The outcome distribution is also unequal, with substantially more recovered patients than deaths. In addition, removing records with missing values can reduce the representativeness of the final dataset if missingness in a larger clinical setting is not completely random. Stronger evaluation would include an external hospital dataset, temporal validation on later COVID-19 waves, calibration analysis, confidence intervals, and subgroup testing by age and sex. Comparing the model with logistic regression, Random Forest and gradient-boosting methods on the same held-out population would also clarify the practical value of CART.

(c) Real-World Applicability

The technique could be used as a clinical decision-support component for prioritising COVID-19 patients who require closer monitoring or rapid intervention. Its strongest practical advantage is that the final tree can be translated into understandable rules using a small set of routinely measured variables such as age, BUN and SGOT. However, deployment in hospitals would require validation on local populations, reliable electronic health-record data, consistent laboratory measurement, and mechanisms for handling missing values. The model should support clinicians rather than replace them, because mortality prediction can affect urgent treatment decisions. Ethical issues include fairness across demographic groups, patient privacy, informed consent and the risk of over-relying on historical clinical data. The model would also need recalibration as treatment protocols, vaccination coverage and circulating variants change.

Task 4: Reflection & Learning Outcome

(a) Three Key Insights Linked to CO4

- Decision trees are valuable because they make the learned classification process visible. The CART example showed how a path through age, BUN and SGOT can be translated into an estimated mortality risk. This connects directly to the CO4 topic of Decision Tree Algorithms and demonstrates why tree models are often described as interpretable or white-box models.
- Feature selection can improve both efficiency and interpretability. Random Forest was used to rank 25 candidate variables before the tree models were trained, reducing the final decision process to six important features. This connects to Inductive Learning because the model learns which attributes provide useful evidence for generalising from labelled patient records.
- Evaluation metrics must match the application. In mortality prediction, recall is particularly important because a false negative may mean that a high-risk patient is not prioritised. The study's CART test recall

of 0.955 compared with accuracy of 0.7824 shows why CO4 model evaluation should consider multiple measures such as accuracy, recall, F1-score and AUC rather than relying on one number.

(b) Proposed Extension: Novel Experiment

A useful extension would be to develop a temporal and external-validation experiment. The existing CART model could be trained using the earlier hospital records and then tested on patients admitted during a later period or at a different hospital. The experiment could additionally compare CART with Random Forest and XGBoost while preserving the same six selected features. This is feasible because the study already identifies a compact feature set and a clear evaluation pipeline. The expected impact would be to determine whether the decision rules remain reliable when treatment protocols, patient demographics and disease conditions change. A calibration curve and subgroup analysis by age and sex could be added to assess whether predicted risk is reliable and equitable across patient groups.

Note on Assignment Scope

This review follows the structure of the CO4 Assessment Tool 2 brief and uses one peer-reviewed article published after 2019 from Springer Nature. The selected study focuses on Decision Trees and Inductive Learning through an applied classification problem. The discussion has been written in an original form while keeping the required four-task structure: article summary, methodology analysis, results and critical evaluation, and reflection with a proposed extension.